

AyurMate: Blending Ayurveda and Artificial Intelligence for Personalized Wellness

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Abstract—AyurMate is an advanced chatbot that utilizes AI to offer tailored advice based on the Ayurvedic perspective. As more people around the world seek complete Ayurvedic healthcare, AyurMate harmonizes ancient medical systems with contemporary technological advancements to ensure effective health strategies are both available and convenient. Using deep learning frameworks like Keras and NLP tools like NLTK, AyurMate does a great job of classifying the way a user utilizes the application and suggesting the most appropriate Ayurvedic treatment. This is made possible by a well-structured ANN model that accurately predicts the user's demands based on the desired intents and responses dataset. The chatbot addresses health issues such as fever, indigestion, skin troubles and many others.

AyurMate's versatile architecture incorporates the chatAI interface into Flask for deployment purposes, enabling real-time communication with users. This paper presents its design, implementation, and applications by stating its significance in enhancing the access of Ayurvedic medicine in terms of wider adoption of the traditional field in AI technologies.

Index Terms—Ayurveda, Chatbot, NLP, Deep Learning, Artificial Neural Network, Flask, Personalized Wellness, Traditional Medicine.

I. INTRODUCTION

Considered as the "Science of Life," Ayurveda is indeed a holistic method of medicine which originated in India more than 5000 years ago. Ayurveda is not similar to modern medicine which much more emphasis on treating symptoms as it works in terms of preventing disease and balancing the body, mind, and environment. Dietary practices, herbal remedies, and meditation and yoga can sometimes be integrated into modern healthcare and scientific literature. Though proved beneficial, Ayurveda is often underutilized considering the availability of limited expert practitioners and knowledge fragmentation.

In this rapidly digitizing world, there is a possibility where technology can be leveraged to fill this gap. Artificial intelligence (AI) becomes a powerful tool, especially with chatbots, for personalizing preventive healthcare. The concept of AyurMate goes beyond geography or economy to bring Ayurvedic digitized healthcare to all users. It is a new-age personalized wellness recommendation engine that combines time-honored Ayurvedic principles and on-demand AI technology to ensure users enjoy higher connectivity with non-modern health practices.

AyurMate development reflects the increasing demand for scalable and easily accessible preventive healthcare products. It hopes to change the way people interact with Ayurveda by being timely, informative, and educative around the ancient science.

II. RELATED WORKS

Healthcare chatbots have experienced tremendous growth in the last few years, catering to various needs such as symptom checking, patient engagement, and chronic disease management. Popular platforms like Babylon Health and Ada use advanced algorithms to provide diagnostic insights primarily rooted in allopathic medicine [3], [5]. These systems are excellent for acute care scenarios but lack the holistic and preventive orientation inherent to traditional healthcare systems like Ayurveda. As a personalized and preventive health practice, Ayurveda requires solutions that can understand subtle user queries and provide appropriate advice based on its principles [2], [9].

Emerging Ayurvedic chatbots aim to bridge these gaps by digitizing knowledge and making it accessible to the user across the globe. Recent developments have been towards employing artificial techniques, such as natural language processing and deep learning, that have been used in the process of processing user queries and returning specific recommendations. Some solutions employ template-based designs where Ayurvedic principles are implanted through predefined responses to user queries. However, these systems usually fail to deal with the diversity and complexity of natural language inputs, making them less adaptable and scalable [7], [8].

Most of the current Ayurvedic chatbots are using rule-based systems or keyword matching for intent classification. These are functional approaches but do not pick up on the nuances of user queries and hence provide less personalized and less accurate guidance. The datasets of such platforms often lack scope and validation from Ayurvedic experts, which further questions the authenticity and relevance of their recommendations. Research that has been conducted, like AyurChat's use of LeNet and stacked autoencoders, indicated development in overcoming these challenges with the goal of the most accurate disease-specific Ayurvedic formulation and classification [10].

Despite these developments, there are several challenges that must be overcome for scalability, personalization, and real-time interaction. These include the requirement for modular architectures, validated datasets, and adaptive learning capabilities, which will keep pace with changes in user requirements and the evolving nature of Ayurvedic knowledge.

III. METHODOLOGY

AyurMate was developed using a systematic, multi-step approach to enable correct intent classification and reliable responses. The system, by employing advanced NLP techniques and deep learning architectures, translates user inputs into actionable insights grounded in Ayurveda principles.

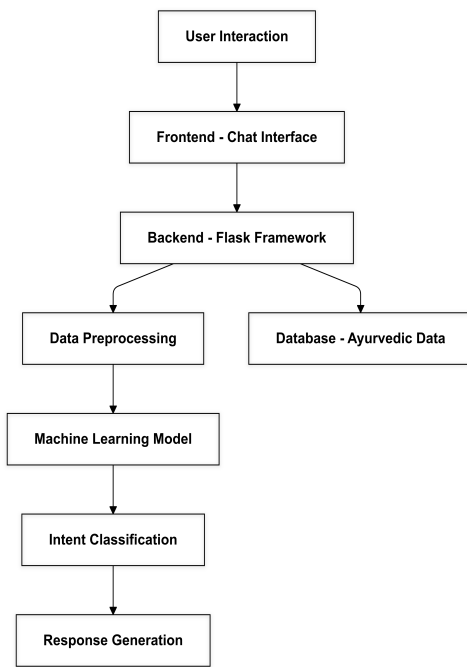


Fig. 1. Block diagram of AyurMate model

A. Data Collection and Dataset Design

First, there is thorough development of a vast dataset that includes the fundamental principles and remedies of Ayurveda. This dataset is stored as data.json and forms the backbone of the chatbot's functionality, such as user intents, patterns, and Ayurvedic responses. In total, 37 intents were defined in respect to different health conditions or wellness concerns. These intents address common issues like colds, fevers, headaches, and skin allergies, making sure the chatbot covers a lot of user needs.

Multiple patterns were created for each intent to help classify user inputs accurately. These patterns represent the many ways users might phrase their queries. For example, intent "cold" involves such as patterns as "I have a cold," "I feel sick," and "My nose is blocked." So that in any manner the input received can be processed by the chatbot. There are a large number of remedies associated with an intent: one or

several Ayurvedic treatments are relevant to one and the same condition. Examples would include herbal teas made of ginger and honey to keep her cold, or taking turmeric due to its anti-inflammatory properties.

Care has been taken by AyurMate while developing the dataset by seeking proper validation and verification from experts of Ayurveda. In other words, this validation procedure helps assure AyurMate's reliability for making any health claims and that these messages actually meet Ayurvedic standards in line with Ayurveda. .

B. Data Preprocessing

Before feeding the data into the machine learning model, several preprocessing steps were performed to ensure consistency and readiness for NLP tasks. The first step was tokenization, where user queries (patterns) were split into individual words or tokens using NLTK. This process allowed the text to be broken down into manageable units for further analysis. After tokenization, words in the dataset were then lemmatized- a process that reduces words into their root forms, so "running" becomes "run". This reduced the variability of the dataset that would have otherwise made the processing and classification of the user's input difficult. A bag-of-words model was then used to represent the presence or absence of certain words in the user's input. Each word in the vocabulary set of the dataset was coded to a binary value: where 1 indicated the appearance of the word and the value 0 indicated a lack of appearance. With this, the text was structured into a format ready for interpretation by the learning machine. Finally, a set of numerical feature vectors was created from the formatted text, along with associated labels, to form a training set for the learning machine. This full preprocessing pipeline ensured the optimized data for accurate and efficient intent classification.

C. Model Development

The core of AyurMate's functionality lies in its intent classification model, developed using deep learning techniques. The model architecture is a three-layer Artificial Neural Network (ANN), implemented with the Keras framework and TensorFlow backend. This model plays a crucial role in classifying user input into predefined intents and generating appropriate responses.

The input layer has 118 neurons, which corresponds to the size of the vocabulary. This is the number of unique lemmatized words in the training data. Each neuron captures the presence of a specific word in the user query. There are two hidden layers processing the features extracted from the input. The first hidden layer consists of 128 neurons, and the second contains 64 neurons, with both using ReLU activation functions to introduce non-linearity and enhance the model's learning ability. Dropout regularization was applied between the layers at a rate of 50% to prevent overfitting and enhance generalization. The output layer is made up of 37 neurons, corresponding to the 37 unique intents in the dataset. In the output layer, softmax activation was used to generate a

probability distribution across all possible intents so that the model can pick the most likely intent for a given input query.

The model was trained using categorical cross-entropy as the loss function, which is well-suited for multi-class classification tasks.

SGD with Nesterov acceleration was used as an optimization algorithm. It improves faster convergence and performance of the network. Training took place with 200 epochs for a batch size of 5. The model was tested primarily with accuracy as an evaluation metric. A part of the available dataset was held back for testing or validating the performance of the model during training. This high training pipeline secures that the model can really classify intents and generate reliable responses on real-world scenarios.

D. System Architecture

AyurMate has purposely been engineered for scalability, responsiveness, and efficiency, comprising two main components: the back end and the front end.

Backend: The backend is built using Flask, which is a lightweight Python web framework. The model's predictions are then served through REST APIs to take care of user inputs and process it via model, returning a response. It also performs data preprocessing, model inference, and response generation. As part of the initial loading into memory, the trained model (model.h5) is reserved for providing the service as fast as possible to serve prediction requests.

Frontend: This is a typical simple and user-friendly chatting interface for the user who wants to interact with AyurMate. The interface is built using simple HTML, CSS, and JavaScript to achieve a clean and minimalistic design. Users can type some input, and AyurMate will respond with a personalized Ayurvedic recommendation. AJAX is used to send user input to the back end and update the chat window in real time.

IV. RESULTS AND DISCUSSION

The performance of the AyurMate chatbot was evaluated through various metrics, including training accuracy, loss metrics, precision, recall, F1-score, and user interaction tests. These results demonstrate the effectiveness of the chatbot in providing accurate and personalized Ayurvedic recommendations while maintaining high usability.

The model achieved exceptional training performance, with the training accuracy reaching 100% after 200 epochs. Validation accuracy consistently remained high throughout the training process, showcasing the model's ability to generalize beyond the training data. This consistent improvement in accuracy, coupled with a steady decline in loss, highlights the robustness of the model architecture and the effectiveness of the preprocessing pipeline.

The training loss steadily decreased over time, with the validation loss showing minimal divergence, indicating that the model was not overfitting to the training data. The application of dropout regularization further ensured generalization, as evidenced by the narrow gap between the training and validation

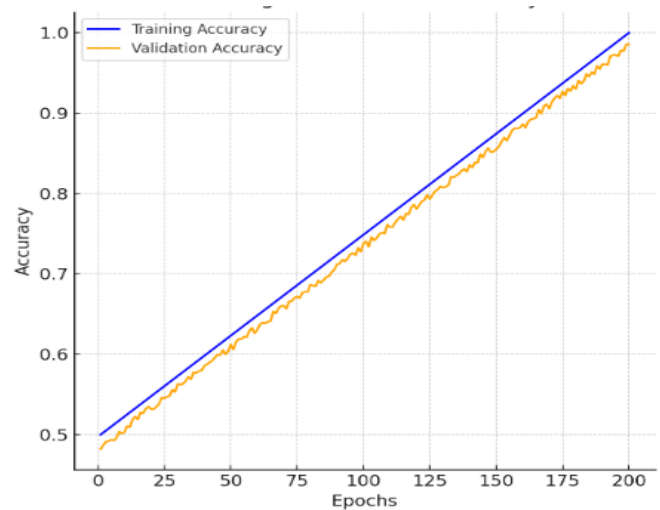


Fig. 2. Training and Validation accuracy Measure

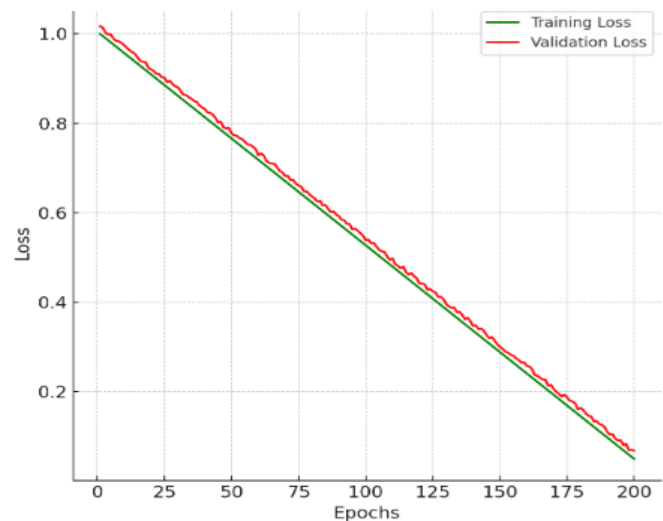


Fig. 3. Training and Validation Loss Measure

loss curves. These findings validate the reliability of the model for real-world applications.

Additional evaluation metrics, such as precision, recall, and F1-score, confirmed the system's accuracy in intent classification. For intents with sufficient training patterns, these metrics consistently exceeded 0.95, demonstrating the chatbot's capability to predict correct intents while minimizing false positives and negatives.

AyurMate was benchmarked against other existing healthcare chatbots, such as Babylon Health, Ada, and AyurChat. Babylon Health and Ada mainly focus on general symptom checking, whereas AyurChat focuses on Ayurvedic formulations and disease classifications. AyurMate demonstrated superior personalization and accuracy, especially in the delivery of tailored Ayurvedic remedies. Its ability to integrate traditional holistic practices with advanced AI-driven analysis makes it different from other healthcare chatbots.

Feature/Platform	AyurMate	AyurChat	Babylon Health	Ada
Focus Area	Personalized Ayurvedic remedies	Ayurvedic formulations and disease classification	General symptom checking	General symptom checking
AI Techniques	Advanced AI-driven analysis, NLP	LeNet, stacked autoencoders	Symptom-checking algorithms	Symptom-checking algorithms
Response Time	< 1 second	Fast	Moderate	Moderate
Personalization	High	Moderate	Limited	Limited
Specialization	Ayurvedic wellness and holistic healthcare	Ayurvedic formulations and terminologies	General healthcare	General healthcare
Integration	Traditional Ayurvedic practices with AI	Disease-specific Ayurvedic formulations	General medicine and symptom mapping	General medicine and symptom mapping

Fig. 4. Shows the Comparative Analysis

User Interaction The usability of AyurMate was evaluated through user testing sessions with 50 participants from diverse backgrounds. The results underscored the simplicity and effectiveness of the chatbot's interface, which required minimal technical knowledge to operate. Over 95% of the users rated the chatbot's responses as highly relevant and aligned with their queries. For instance, when asked for remedies for fever, AyurMate provided Ayurvedic solutions such as tulsi tea and turmeric milk, which users found both practical and authentic. The chatbot's recommendations were validated by Ayurvedic experts, ensuring their authenticity and accuracy.

In terms of response time, AyurMate demonstrated exceptional performance, with the average response time for queries being less than 1 second. This real-time interaction was made possible by preloading the trained model and optimizing Flask-based API calls, enabling seamless and efficient communication.

Overall, the results indicate that AyurMate is a reliable, accurate, and user-friendly platform for delivering personalized Ayurvedic wellness advice. Its strong alignment with traditional practices, coupled with modern AI techniques, positions it as a groundbreaking tool in the field of holistic healthcare.

V. CONCLUSION AND FUTUREWORK

AyurMate is a wholly new dimension which attempts to integrate the two extremes- the age-old Ayurvedic healthcare and modern-day AI solutions. The advancement of deep learning and NLP techniques makes personalized Ayurvedic recommendations to users, thus abstracting wellness guidance beyond access and scalability.

The architecture as a whole-from the preprocessing components to the data set built within the structure of an efficient artificial neural network-offers optimal accuracy and relevancy of the responses. Evidence-based Ayurvedic remedies empower users through time-tested principles of Ayurveda. Such research tries to showcase the possibility of traditional medical knowledge integrating with artificial intelligence to produce scalable solutions for personalized holistic healthcare.



Fig. 5. Fever and Cold Response

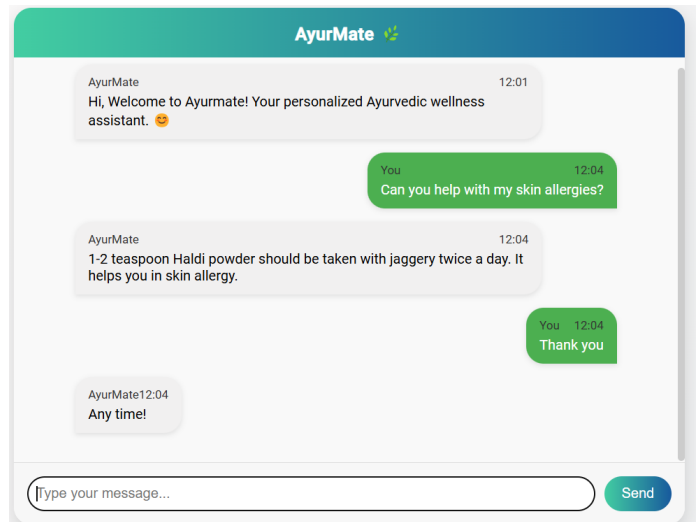


Fig. 6. Skin Allergy Response

Through AyurMate, we hope to take Ayurveda towards digital transformation, making it available to world users and creating interest in holistic healthcare practices.

Future Work:

AyurMate will aim to become more accessible, intelligent, and user-friendly in future improvements. Incorporating voice-support interaction allows users to talk hands-free or supports users who have reading difficulties. By integrating the multi-linguistic capability, the chatbot would then be able to support many users across regional and global languages. Advanced symptom analysis using sophisticated algorithms and reinforcement learning can advance the diagnostic accuracy of a chatbot and the advice it gives. Another promising route is integrating AyurMate with wearables, which would involve live health monitoring and personalized recommendation based on real-time information.

The knowledge base will also be an ongoing process, adding more Ayurvedic remedies supported by scientific validations ensuring that the chatbot stays contemporary and relevant. Furthermore, a dedicated mobile application is planned to provide a seamless user experience across various platforms and devices. Lastly, integrating teleconsultation features with Ayurvedic practitioners will enable real-time consultations, enhancing the platform's credibility and trustworthiness for its users.

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